

Pokhara University
Faculty of Science and Technology

Course Code: ELX 120

Course title: Electronic Devices and Circuits (3-1-2)

Nature of the course: Theory and Practical

Level: Bachelor

Full Marks: 100

Pass Marks: 45

Total Lectures: 45 hours

Program: BE

1. Course Description

The purpose of the course is to provide the fundamental concept of semiconductor and their application in electronics devices. Furthermore, this course provides the idea of analyzing and designing electronic device and circuits.

2. General Objectives

The course is designed with the following general objectives:

- To provide the concept of semiconductor diode and their application in dc power supply, clipper and clamping circuits
- To familiarize the students with construction, working principle, and characteristics of different types of transistors and their application.
- To make the students competent in analyzing and designing the different electronic circuits.

3. Methods of Instruction

Lecture, Tutorial, Discussion, Readings and Practical works

4. Course in Detail

Specific Objectives	Contents
• Explain the basic concept of p-n junction and their implementation on diode and Zener diode.	Unit I: Semiconductor Diode (5 hrs) 1.1 Introduction to insulator, semiconductor and conductors 1.2 Conduction principle in semiconductors, electrons and holes, donor and acceptor impurities, n-and p-type semiconductors 1.3 p-n junction: formation of space-charge region, energy band structure and barrier potential 1.4 The p-n junction under forward bias and reverse bias, characteristic curve and temperature effect 1.5 Junction breakdown, Junction and diffusion capacitances 1.6 Diode switching times 1.7 Construction, characteristics and



	applications of Zener diode
- Explain the importance of bipolar transistors in modern electronic circuits - Describe the construction, working principle and characteristics of bipolar transistors - Discuss the DC/AC load line and Q-point in BJT and analyze the different method of BJT biasing circuits	Unit II: Bi-polar Junction Transistor (6 hrs) 2.1 Introduction of bipolar junction transistor (BJT) 2.2 Construction and working principle of different types of BJT, Mode and configuration in BJT 2.3 Current gain of BJT at different configuration and their relationship 2.4 Input and output characteristics of transistor 2.5 Comparison of CB, CE and CC configurations 2.6 Biasing and different methods of biasing, DC/AC load line and Q-point in BJT 2.7 The transistor as an amplifier and switch, BJT switching times 2.8 Concept of bias stabilization and stability factor
- Explain and analyze the different types of rectifiers - Discuss and analyze filter, and regulators circuit - Explain and analyze the operation of clipping and clamping circuits	Unit III: DC Power Supply (5 hrs) 3.1 Rectifier and their types, average value, RMS value, ripple factor, efficiency 3.2 Filtering process and Shunt capacitor 3.3 Clipper and clamping circuits 3.4 Power supply: Regulated and unregulated power supplies 3.5 Transistor series and transistor shunt regulators
- Describe the construction, working principle and characteristics of FET and MOSFET - Discuss and analyze the DC load line of different FET biasing circuits - Visualize the concept of small signal model in JFET	Unit IV: The Field Effect Transistor (FET) (4 hrs) 4.1 Comparison between FET and BJT 4.2 Construction and working principle of JFET 4.3 Drain and transfer characteristics of JFET and JFET parameters 4.4 Biasing circuits and DC load line 4.5 Construction and working principles of DMOSFET and EMOSFET
- Explain the concept of AC analysis of BJT - Apply the small signal low frequency analysis models using r_e model of transistor in different transistor configurations	Unit V: The Small Signal Low Frequency Analysis Model of BJT (6 hrs) 5.1 Introduction and basic concept of AC analysis of BJT 5.2 r_e model of BJT, amplifier configuration and their expression for voltage gain, current gain, input impedance and output



	impedance using r_e model 5.3 Analysis of transistor amplifier circuit using r_e model 5.4 Emitter follower
- Visualize the concept of multistage amplifier and various coupling methods. - Analyze multistage amplifier in term of voltage gains, Current gains, input and output impedance using r_e model - Explain the Darlington-pair amplifier and calculate its effective beta	Unit VI: Multistage Amplifiers (4 hrs) 6.1 Multistage amplifier and its importance, methods of coupling 6.2 Gain calculation of n-stages cascaded amplifiers 6.3 Expression of voltage gains, current gains, input and output impedance for two stages RC coupled amplifier using r_e model 6.4 Choice of configuration in a cascade 6.5 Darlington-pair amplifier and its effective beta
- Define the concept of large signal amplifier - Explain the operating principle and analyze the power conversion capabilities of different classes of Amplifiers with its merits and demerits - Visualize the concept of push-pull amplifiers and transformer coupled push-pull amplifier	Unit VII: Large Signal Amplifiers (4 hrs) 7.1. Large signal amplifier and its importance 7.2. Analysis of Class A, B, and AB amplifier 7.3. Push-pull amplifiers and Transformer coupled push-pull amplifier 7.4. Cross over distortion 7.5. Amplifier efficiency, power dissipation and heat sinks
- Explain negative and positive feedback amplifier with its importance - Design RC and LC oscillator using Op-amp	Unit VIII: Feedback Amplifiers (7 hrs) 8.1. Classification of feedback amplifier 8.2. Negative feedback amplifiers and advantages of negative feedback (gain, stability, extension of bandwidth, signal to noise ratio, input and output impedances) 8.3. Importance of positive feedback on oscillation 8.4. Barkhausen criteria for oscillation 8.5. RC and LC oscillator using Op-amp 8.6. Multivibrators: Astable and monostable



- Visualize the operational amplifiers and its properties - Explain type of amplifier (Inverting and non-inverting) and apply its application in operational amplifier.	Unit IX: Operational Amplifier (4 hrs) 9.1. Basic Model 9.2. Ideal and non-ideal properties 9.3. Virtual ground concept and CMRR 9.4. Inverting and non-inverting amplifier 9.5. Summing, Integrator, and differentiator amplifier and their applications
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5. List of Tutorials

The following tutorial activities of 15 hours per group of maximum 24 students should be conducted to cover all the required contents of this course:

S.N.	Tutorials
1	Numerical related to Zener diode as a voltage regular
2	Numerical related to current gain, biasing circuits, DC/AC load line and Q point in BJT.
3	Numerical related to clipper and clamping circuits.
4	Numerical related to biasing circuits (DC load line, Q point) of JFET.
5	Numerical related to r_e modeling of different BJT amplifier configuration to calculate voltage gain, current gain, input impedance and output impedance.
6	Numerical related to calculation of voltage gain, current gain, input impedance and output impedance for two stages RC coupled amplifier using r_e model.
7	Numerical related to analysis of Class A, B, and AB amplifiers.
8	Numerical related to negative feedback amplifiers and design of RC and LC oscillator using Op-amp.
9	Numerical problems related to design of operational amplifier.

6. Practical Works (30 hours for a group of maximum 24 students)

S.N.	Practical Works
1	Familiarization with different basic components and measuring tools.
2	Study of V-I Characteristics of different PN junction diodes
3	Study of rectification characteristics of half-wave and full-wave rectifier.
4	Study of clipper and clamper circuits.
5	Study of Zener diode characteristic as voltage regulator
6	Study of input and output characteristics of CE and CB transistor amplifier.
7	Study of drain and transfer characteristics of JFET
8	Study of drain and transfer characteristics of MOSFET
9	Measurement of gain in single stage and multistage amplifiers.
10	Measurement of efficiency of class A and Class B push pull power amplifiers.
11	Design of RC and LC oscillator using Op-amp and their verification.
12	Measurement of regulation in series regulator against change in input voltage and load resistance.



7. Evaluation System and Students' Responsibilities

Evaluation System

In addition to the formal exam(s), the internal evaluation of a student may consist of quizzes, assignments, lab reports, projects and class participation. The tabular presentation of the internal evaluation is as follows.

Internal Evaluation	Weight	Marks	External Evaluation	Marks
Theory		30	Semester-End examination	50
Attendance & Class Participation	10%			
Assignments	20%			
Presentations/Quizzes	10%			
Internal Assessment	60%			
Practical		20		
Attendance & Class Participation	10%			
Lab Report/Project Report	20%			
Practical Exam/Project Work	40%			
Viva	30%			
Total Internal		50		
Full Marks: 50 + 50 = 100				

Student's Responsibility

Each student must secure at least 45% marks separately in internal assessment and practical evaluation with 80% attendance in the class in order to appear in the Semester End Examination. Failing to get such score will be given NOT QUALIFIED (NQ) to appear the Semester-End Examinations. Students are advised to attend all the classes, formal exam, test, etc. and complete all the assignments within the specified time period. Students are required to complete all the requirements defined for the completion of the course.

8. Prescribed Books and References

Text Books

1. Bogart, T. F. Beasley, J. S. & Rico, G. (2004). *Electronic devices and circuits* (6th ed.). Pearson Education
2. Boylestad, R. & Nashelsky, L (2012) *Electronic devices and circuit theory* (11th ed.). Prentice Hall, India.

References

1. Sedra, A. S. & Smith, K. C. (2004). *Microelectronic circuits* (5th ed.). Oxford University Press, New York.
2. Millman, J. Halkias, C. C. & Jit, S. (2010). *Milman's electronics devices and circuits* (3rd ed.). Tata McGraw Hill Education.

